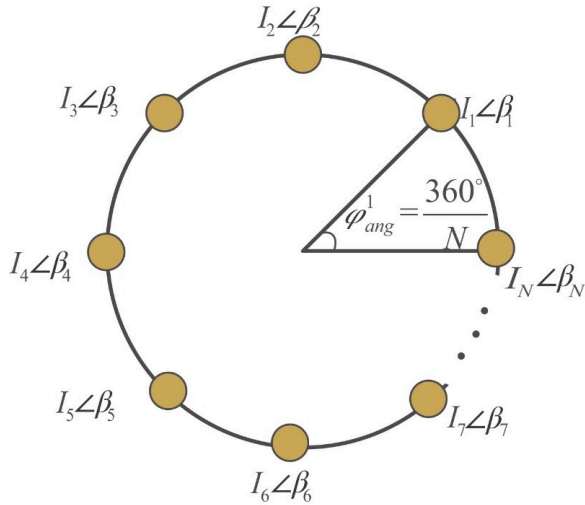


Optimize the circular antenna array using the PSO algorithm

The circular antenna array is a frequently used antenna array for wireless communication systems. The topology of the circular antenna array is shown as follows. The objective function consists of the side lobes, the directivity, the desired maxima φ , and the null control.



Assuming you are an antenna array designer, you are required to optimize the feeding amplitude and phase of a 5-element circular array. There are 10 design variables, including two types: (1) amplitude, with a search range of [0.1, 1], and (2) phase, with a search range of [-180, 180].

The objective function for the array optimization (antennafunccircular.m, see reading materials) is provided in the attached files. When the objective function value is less than -10, it indicates that the circular array has a good performance.

Particle swarm optimization (PSO) is a good choice for this task. The recommendations are as follows: the swarm size should not exceed 200, and the number of optimization iterations should not exceed 200.

Script

Save

Reset

MATLAB Documentation (https://www.mathworks.com/help/)

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```
1 %This function calculates the fitness value of array 'x1' which is returned in 'y'
2
3
4 null=0;
5 phi_desired=180;
6 distance=0.5;
7
8
9 pi=3.141592654;
10 dim=length(x1);
11 y=0;
12 [temp num_null]=size(null);
13 num1=300;
14 phi=linspace(0,360,num1);
15 phizero=0;
16 yax(1)=array_factorcir(x1,(pi/180)*phi(1),phi_desired,distance,dim);
17 maxi=yax;
18 phi_ref=1;
```

Run Script

Tests: All Tests Passed

Submit

- ✔ Is the number of generations too large?<=200
- ✔ Is the swarm size too large?
- ✔ Has particleswarm been used?
- ✔ Was the result correct?

Output

Optimization ended: number of iterations exceeded OPTIONS.MaxIterations.

